NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE | midori lab

OurMethod: Your Paper Title Goes Here in Two Balanced Lines

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Abstract

Template text: every claim, name and number in this file is a placeholder. Large models often fail when **two input sources disagree**, even though the evidence needed for the answer is still present in one of them. We propose **OurMethod**, a *training-free* method that detects such disagreements and corrects the joint prediction at inference time. On BenchA and BenchB, **OurMethod** improves overall accuracy on Model-7B from 70.5% to 85.9% in this illustrative setting.

Date: October 17, 2026
Code: https://github.com/your-org/your-repo
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1 Introduction

Background paragraphs stay in plain text. Recent models jointly process several input sources and are widely used in downstream applications (Vaswani et al., 2017; Goodfellow et al., 2016). Combining sources, however, does not guarantee that the answer is grounded in the right one.

The key bottleneck is that **the relevant evidence is often still recoverable**, yet the joint prediction ignores it. The two sources differ in *which question they can answer*, while sharing the same *internal representation*. This raises the central question of this work:

> *Can a model recover the answer from the relevant source when the joint prediction has already gone wrong?*

We answer it with **OurMethod** (Figure 1), which first locates the relevant source with a ✿ ROUTER and then steers the joint representation with a ❀ CORRECTOR. The main contributions are:

❶ ***A recoverable failure mode.*** We identify cases where the joint prediction is wrong but a single-source view is right, and measure how often they occur.

❷ ***A training-free correction.*** **OurMethod** detects these cases and corrects them at inference time without updating model weights.

❸ ***Controlled evaluation.*** We evaluate on BenchA and BenchB with two model sizes and compare against PriorA and PriorB.

2 Related Work

Joint multi-source models. Plain text for the survey of one category of prior work (Hinton et al., 2006). End with one sentence that positions this paper against the category.

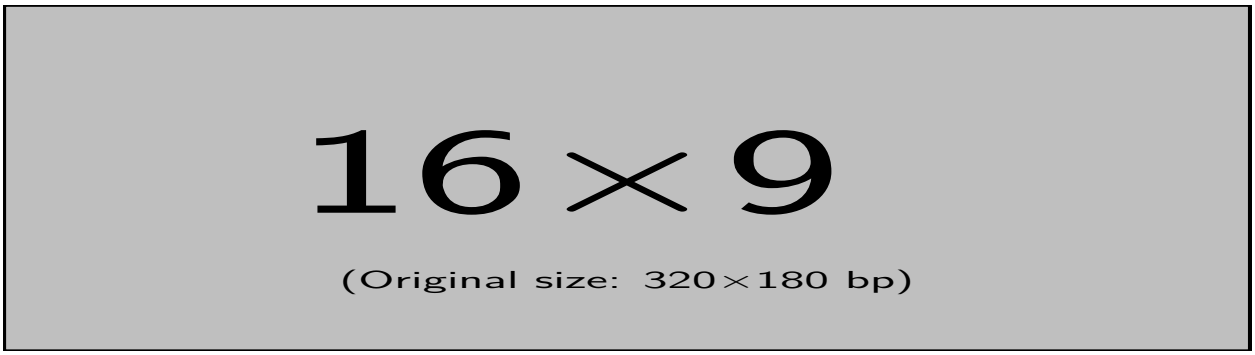


Figure 1 **Overview of `OurMethod`.** ***(a) Detect:*** the ✿ ROUTER selects the relevant source and checks for a decision conflict. ***(b) Correct:*** the ❀ CORRECTOR steers the joint representation toward that source. Flower markers match the module names in the text.

Algorithm 1 **`OurMethod`** inference

Require: sources a, v; question q; strength α
$\hat{y} \leftarrow \arg\max_c z_c(a, v, q)$ ▷ joint prediction
$\hat{y}_{\text{uni}} \leftarrow \text{ROUTER}(a, v, q)$ ▷ Stage 1: Detect
if $\hat{y}_{\text{uni}} \neq \hat{y}$ **then**
  apply Equation (2) at the selected layers ▷ Stage 2: Correct
  $\hat{y} \leftarrow \arg\max_c z'_c$
end if
return $\hat{y}$

Inference-time correction. Methods such as PriorA and PriorB adjust the output distribution at decoding time. They *suppress* errors, whereas **`OurMethod`** *recovers* the missing evidence.

3 Method

Problem setup. Given input sources a and v and a question q, a model with parameters θ produces a score z_c for every candidate $c \in \mathcal{C}$. The prediction and its margin are

$$\hat{y} = \arg\max_{c \in \mathcal{C}} z_c, \qquad m = z_{\hat{y}} - \max_{c \neq \hat{y}} z_c. \tag{1}$$

▷ **Stage 1: Detect.** The ✿ ROUTER withholds the competing source and predicts $\hat{y}_{\text{uni}}$. A conflict is flagged when $\hat{y}_{\text{uni}} \neq \hat{y}$.

▷ **Stage 2: Correct.** For flagged inputs, the ❀ CORRECTOR shifts the hidden state $\mathbf{h}_\ell \in \mathbb{R}^d$ at layer ℓ:

$$\mathbf{h}'_\ell = \mathbf{h}_\ell + \underbrace{\alpha\,(\mathbf{h}_\ell^{\text{uni}} - \mathbf{h}_\ell)}_{\text{CORRECTOR residual}}, \tag{2}$$

where α scales with the confidence of the single-source view. Algorithm 1 summarizes both stages.

4 Experiments

Setup. We evaluate `Model-3B` and `Model-7B` on `BenchA` and `BenchB`; the hyperparameters are tuned on a held-out split of TrainCorpus. Baselines are PriorA and PriorB under the same decoding settings.

Observation ❶: `OurMethod` gives the largest gain on `BenchB`. Relative to the strongest baseline in each column, accuracy rises by 6.5 points on `BenchA` (75.5 → 82.0) and by 14.3 points on `BenchB` (75.5 → 89.8), as shown in Table 1.

Table 1 Main results (%). Placeholder numbers on `Model-7B`. Higher is better for all columns. Within each column, the best value is **bold** and the second best is underlined; the shaded row marks our method and does not by itself indicate the best result.

Backbone	Method	BenchA		BenchB		Overall Acc.
		Acc.	F1	Acc.	F1	
`Model-7B`	Baseline	69.3	75.7	71.8	77.0	70.5
	+PriorA	75.5	79.4	66.5	73.5	71.0
	+PriorB	72.1	77.8	75.5	79.1	73.8
	+OurMethod	**82.0**	**84.1**	**89.8**	**89.8**	**85.9**

Observation ❷: Across the two sizes we test, the gain is larger on `Model-7B`. Table 2 shows a gain of 5.6 points on `Model-3B` and 15.4 points on `Model-7B`. With only two sizes, this does not establish a trend with scale. Plain text continues the analysis here. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Table 2 Scaling. Overall accuracy (%); gains over Base in parentheses.

Model	Method	Acc.
`Model-3B`	Base	76.9
	↪ **+OurMethod**	82.5 (+5.6)
`Model-7B`	Base	70.5
	↪ **+OurMethod**	85.9 (+15.4)

Takeaway

The overall gain is not uniform across benchmarks or model sizes. Report `BenchA` and `BenchB` separately, as in observations 1 and 2.

5 Conclusion

Summarize the findings in plain text, reusing the entity macros.

Limitations and Future Work. `OurMethod` assumes that **the relevant evidence is still present in one source**; it may help less when that evidence is weakly encoded.

AI use statement

Describe how generative AI tools were used in preparing the manuscript.

Reproducibility Statement

Point to the sections and appendices that support reproduction.

References

Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep Learning*. MIT Press, 2016.

Geoffrey E. Hinton, Simon Osindero, and Yee Whye Teh. A fast learning algorithm for deep belief nets. *Neural Computation*, 18:1527–1554, 2006.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, 2017.

Appendix

A Implementation Details

A.1 Prompt for the ROUTER

Prompts and other raw materials go in a breakable monospace box; the text is kept verbatim.

Router prompt

```
SYSTEM: You will see a question about an audio-visual clip.
USER: {question}
To answer this question, which modality is needed (audio, video, or both):
```

B Typography Key

Table 3 **Formatting roles used in this template.** Delete this appendix in your paper.

Role	Macro	Example
Our method	`\rpsMethod`	**`OurMethod`**
External method	`\rpsBaseline`	PriorA
Model	`\rpsModel`	`Model-7B`
Dataset	`\rpsDataset`	TrainCorpus
Benchmark	`\rpsBenchmark`	`BenchA`
Named module	`\rpsComponent`	✿ ROUTER, ❀ CORRECTOR
Method stage	`\rpsStage`	▷ **Stage 1: Detect.**
Key phrase	`\rpsStrong`	**bottleneck**
Contrast	`\rpsTerm`	*suppress* vs. *recover*
Contribution head	`\rpsStrongTerm`	***Short title.***
Finding	`\rpsObservation`	**Observation ❶:** …
Gain / drop	`\gain`, `\drop`	85.9 (+15.4), 70.1 (−2.3)